Figure 9.9. When electrons are packed together, they prefer to have their spins aligned in the same direction because it is a lower energy configuration than when their spins are anti-aligned, and all physical systems tend toward lower energy (all else being equal).

Figure 9.10. This is a high-resolution Ising model where each pixel represents an electron. The lighter pixels are spin-up, and black is spin-down. You can see that the electrons organize into domains where all the electrons within a domain are aligned, but nearby electrons in an adjacent domain are anti-aligned with respect to the first domain. This organization reduces the energy of the system.

Figure 9.11. This is a depiction of a 2D Ising model of electron spins where + is spin-up and – is spin-down. There is a domain of electrons that are all spin-down (highlighted in black), and these are surrounded by a shell of spin-up electrons.

Figure 9.12. This is a 1D Ising model representing the electron spins for electrons arranged in a single row.

Figure 9.13. The Q function for agent j accepts a parameter vector and a onehot encoded joint-action vector for the neighbors of agent j.

Figure 9.14. A 1D Ising model of electrons arranged in a single row.

Figure 9.15. We are representing the 1D Ising model with a single binary vector, but it is actually a circular grid because we treat the leftmost electron as being immediately next to the rightmost electron.

Figure 9.16. This is a string diagram for the main training loop. For each agent, j, the corresponding Q function accepts a parameter vector and the joint-action vector for agent j, denoted a–j. The Q function outputs a 2-element Q value vector that is input to the policy function, and it chooses an action (a binary digit) that then gets stored in a mirror (clone) of the grid environment. After all agents have chosen actions, the mirrored grid synchronizes with the main grid. The rewards are generated for each agent and are passed to the loss function, which computes a loss and backpropagates the loss into the Q function, and ultimately into the parameter vector for updating.

Figure 9.17. Top: The losses for each agent over the training epochs. You can see that they all decrease and at a minimum at around 30